PAPER 3 — WHAT TO DO NEXT

Your 13 items, reordered by what unblocks what. 24 August 2026. Companion to Handoff v11.0.

THE ONE-LINE VERSION

Data collection is now trustworthy for **straight (0 deg) cylinders**. The next real win is making collection **unattended**, and the thing standing in the way is that closing is open-loop: the gripper squeezes to a hand-picked angle and nothing checks that contact happened. Close the loop, then batch, then scale.

ALREADY DONE (your items 1 and 2)

#	Item	State
1	Grid design is collision-aware (item 1)	**Done.** Per-point check against the measured gripper; refuses rather than emits a colliding grid.
2	Reachability is gripper-vs-object aware (item 2)	**Done.** Checkbox on by default; ledger now reports gripper_modelled: true.

*One caveat carried into the list below: the gripper model contains the body, coupler and knuckles — **never the fingers or pads**, because they must touch the object. Finger-level jams are handled by the grid designer's throat limit instead, not by cuRobo.*

TIER 1 — BEFORE YOU COLLECT THE DATASET

These decide whether the data you collect is trustworthy, or whether you find out a week later that it wasn't.

A. Contact-aware closing (your item 3 + item 4, merged)

Answer to your question: no, the sequential data is not contact-aware. Every grasp in a sweep closes to the same fixed close_rad read from the calibration file — a hand-picked angle per diameter, with a D26 fallback for any size not in the file. Nothing checks that contact actually happened; you only learn a grasp was dead afterwards, from the peaks.

You are right that this is one feature, not two. Closing until a contact target is reached does three things at once: it removes the guessed angle, it guarantees contact by construction, and the pose you stop at IS the offset — so it also replaces the calibration guesswork rather than sitting beside it.

What was checked this session: the collector runs inside Isaac Sim, in the same process as the sensor, so a closed loop needs no streaming, no server and no ROS2 — it can read the signal directly. But it currently talks to the TSF-85 extension through carb settings and only WRITES them; every tactile number comes from the CSV afterwards. So the loop needs a signal the collector does not have yet.

Signal	Available?	Verdict
Geometric indentation	today	Calibrator already reads these prims. Exact in sim, but assumes the object pose and diameter are known — will not transfer to hardware.
Pad deformation, max|sim-rest|	today	**Recommended.** Physical, CNN-free, already live in your logs (0.000000 -> 1.157376 during a close), and the same idea works on a real sensor.
Tactile counts (CNN)	no	Needs the extension to expose its prediction, and inherits the CNN's own open questions.

Build it as an option, not a replacement. Keep the fixed-angle path exactly as it is, add a mode that closes until the target is met, and write pad_offset_calibration_indent.json so the two can be diffed across all nine diameters. Also log the achieved contact per grasp into the ledger — that is the dead-grasp detection, and it comes almost free once the signal is in the loop.

Why it matters beyond convenience: today "firm" only means peak sum above 1000, and that gate has never rejected anything. Your nine entries span 8600–15400 — a 1.8x spread across diameters that are all supposed to be squeezed the same. Deformation or indentation is a physical quantity you can hold constant; tactile counts are not.

Effort: medium. **Blocks:** item B.

B. Unattended batch runner (your item 8)

The biggest multiplier on the list. One command, N configs, N folders, come back a week later.

Shape it as: the GUI writes a numbered set of configs into a batch folder; a runner loops them, launching the collector once per config with its own GRASP_RUN_DIR, appending one result line per run to a batch ledger. It must **survive a failure** — a config that refuses or crashes gets logged and skipped, never allowed to kill the queue.

Two warnings before leaving it running for a week:

- **Disk.** The *_mesh_state.csv and *_deformations.csv files are written every physics frame and are the bulk consumers. Your root partition already filled to 100% once and stalled boot. 100 unattended runs will do it again unless those writers are off or the batch prunes as it goes.
- **Do not batch rolled or tilted configs yet.** Until the diagonal artifact is fixed (item F) those maps are mislabelled at the source, and a week of rolled data is a week wasted.

Effort: medium. **Needs:** A first, so a dead grasp inside run 57 is recorded rather than silent.

TIER 2 — THE ACTUAL PAPER

D. Collect the straight dataset at scale, then train (item 13)

With A–C in place this is the point of everything above. Run the diameter sweep unattended, grow the dataset, and retrain at increasing sizes so you can show the learning curve — which is itself a result, and the honest way to decide when you have enough data.

Expect to come back to collection when training exposes something; that loop is normal and worth budgeting for.

E. Calcul Quebec (item 10)

Start the account now, in parallel with everything else — sponsorship and access have latency you cannot compress later. JP sponsors it. Berith has an open question on whether he has run Isaac Sim there; that answer changes whether you can collect at scale or only train at scale.

Realistically training will move there long before collection does.

F. Diagonal artifact / Berith's new model (items 5 and 11)

Status changed this session: Berith has confirmed the tilt problem is real and is working on it. That downgrades this from "your problem to diagnose" to "a dependency to track", and it removes the pressure to keep probing it yourself.

On your incremental-pressure experiment (item 5): worth doing, but as one bounded afternoon, not a campaign. Take a single grasp and step close_rad in until either the tilt appears or the object visibly deforms, and record the peak sum and blob angle at each step. Two outcomes, both useful: if the angle appears under load it is a contact-area effect and Berith needs to know; if it never appears it confirms his diagnosis independently and you can stop wondering.

> *Do it **after** the calibration work, because indentation in mm is exactly the axis you would be stepping along, and doing it in counts would make the result hard to interpret.*

Until the new model lands, collect 0 deg only. If you want orientation variation before then, 0 and 90 deg are the two the CNN renders faithfully.

G. Live tactile window (your ROS2 question)

Possible, yes — but keep it separate from item A. The control loop needs in-process access and no streaming at all. A live window is a different problem, and only that one needs a stream.

Route	Cost	Note
Tail the CSVs	an afternoon	The extension already writes them continuously. A small viewer polling the file gives both pads live, with zero new infrastructure.
Berith's Qt server	dependency	Already exists, but its 2.5 Hz vs 60 Hz rate is one of his three open questions.
Isaac ROS2 bridge	heaviest	Worth it only if you want one viewer for sim AND the real UR5e, which ran ROS2 Humble in Paper 2.

For it: today's jam was invisible because you run headless — a live map would have shown peak 4 in seconds instead of after a full sweep and a stitch. **Against:** unattended batch means nobody is watching anyway, and a per-run log catches the same thing.

Suggestion: build the closing loop first, since it needs none of this. If you still want the window afterwards, do the CSV-tail version rather than committing to ROS2 before you know you need it.

H. New objects from Berith (item 9)

Valuable — the dataset is entirely cylinders, and Paper 1's whole premise is three shape classes. But sequence it after B, or you will be hand-driving each new shape.

One thing to know before starting: the throat measurement is shape-agnostic — it measures the gripper, so it stays valid. What each new shape needs is its own line of arithmetic in design_grid saying how deep that shape can enter, plus its own contact-band model. A box is easy (flat face, no compliance band). A sphere is harder.

I. Real-robot fork in the GUI (item 12)

Vincent's hedge against Berith's pipeline being in flux, and the collect_real.py path already exists. Its value is that it is **immune to the diagonal artifact** — a real sensor has no CNN in the way — so it can answer the orientation question independently.

Keep it warm rather than parked, but it is not on the critical path while sim collection is finally working.

TIER 3 — ANSWERED, NO ACTION NEEDED

J. Band width "always 8" (item 6)

It is not stuck at 8 — it is working. Your last three runs reported fitted widths of **8.5, 20.5 and 22.0 mm**. The 8.0 is only the fallback default in blob_axis.py, used when the fit checkbox is off or the fit fails.

Two real caveats worth knowing, though: the search range **stops at 22.0 mm**, and your validation run picked exactly 22.0 — so that value is a lower bound, not a measurement. And the fit prints its own large-residual warning on every run so far, meaning the model does not match the maps well at any width. Treat the number as indicative. If you want it to be a measurement, widen the search range past 22 mm — a one-line change.

K. A stitched map from fake blob contacts (item 7)

My honest view: low value, and I would drop it. The stitcher is already validated — two independent runs of the same sweep gave a 2.4% RMS noise floor and visually indistinguishable maps, and the round-trip SSIM/TC checks exist. A synthetic-blob map would mostly re-test the splat geometry, which the coverage panel and the rotated-rectangle check already cover.

There is one version that would earn its place: synthesise the **expected** patch that blob_axis already computes for each pose, stitch those, and compare against the real stitched map. That is a sensor-model test, not a stitcher test, and it would give you a picture of the diagonal artifact accumulated across a whole sweep — something you could hand Berith. Only worth it if he asks for it.

SUGGESTED ORDER, IN ONE PLACE

	Item	Why here	Effort
1	Contact-aware closing (3 + 4)	Removes the guessed angle, guarantees contact, and gives dead-grasp detection almost free	M
2	Unattended batch runner (8)	The biggest multiplier on your time	M
3	Collect at scale + train (13)	The paper	L
4	Calcul Quebec (10)	**Start in parallel now** — access has latency	S then wait
5	Incremental-pressure tilt probe (5)	One bounded afternoon; useful to Berith either way	S
6	Live tactile window (ROS2 question)	Do the CSV-tail version; not needed by the control loop	S–L
7	New objects (9)	Needs the batch runner first	M
8	Real-robot fork (12)	Parallel path, immune to the artifact	M
9	Berith's new model (11)	Track, don't chase	—
10	Band width (6), fake-blob map (7)	Answered below; no action	—

If you only do one thing: items 1 and 2 together. They convert your time from babysitting runs into collecting data, they remove the last piece of hand-tuning from the pipeline, and everything in Tier 2 gets cheaper once they exist.